Knowledge-First Inverse Modelling: Mechanism Discovery over Parameter Optimization

A cardiomyocyte active-stress MPM case study in Plexus

Abstract

Modern differentiable simulators make parameter optimization easy. They do not make mechanism discovery easy. In many inverse-modelling problems the natural objective is to find the parameter vector that best fits the data. This parameter-optimization view treats the simulator as a black box: propose parameters, run the model, compute a score, move toward a better score. It is efficient when the forward model is already correct, but it becomes misleading when the model is incomplete: a better score may only mean the optimizer has found a compensating artifact, not that the model has captured the mechanism. A knowledge-first strategy uses optimization differently. Each batch is not only a step toward a better fit; it is an experiment designed to learn how the system works. Parameters are mechanistic levers, a failed run is useful when it falsifies a hypothesis, and a successful run is useful only when its improvement can be explained in terms of morphology, dynamics, or physical coupling. The objective is therefore not just $\theta^* = \arg\min_\theta \mathcal{L}(\theta)$ but the construction of a map

$$\theta \longrightarrow \text{mechanism} \longrightarrow \text{morphology} \longrightarrow \text{fit}.$$

We make this concrete on the inverse problem of fitting real cardiomyocyte tissue motion with a differentiable active-stress material-point-method (MPM) model built from registered Plexus operators, scored by loop *morphology* (the LoopScore metric) rather than framewise displacement. After many batches the deliverable is not a single optimum but a set of mechanistic constraints on what the forward model must look like: **each mechanistic insight removes an entire class of incorrect models**, and a fit that plateaus despite continued optimization is read not as an optimization failure but as evidence that the forward model is missing a genuine physical ingredient. This closes a discovery loop that we take to be a defining move of the method — *optimization discovers residuals; residuals generate hypotheses; hypotheses become operators; operators extend the language*:

$$\text{residual} \longrightarrow \text{hypothesis} \longrightarrow \text{operator} \longrightarrow \text{extended language}.$$

A residual the *current* operator language cannot break is therefore not a dead end but a prompt: the accumulated knowledge names the missing physical ingredient, and it is added as a new registered operator. The central claim thus has a constructive counterpart — the inverse problem becomes an engine for discovering the operators a correct forward model requires, so that fitting the data and *building the language that can express it* are the same activity.

1 Introduction

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This distinction matters most exactly when optimization stalls. When the score stops improving despite continued training and small architectural changes, the parameter-optimization view sees a disappointing plateau; the knowledge-first view reads the *same* plateau as a measurement — evidence that the forward model is missing a genuine physical ingredient, not that the optimizer has failed. That re-reading is only available if each run has been treated as an experiment: the parameters must be mechanistic levers whose effect on morphology is recorded, the score must measure the object of interest (here loop *shape*, not framewise displacement), and the accumulated knowledge must be organised so that a residual can be attributed to a specific morphological axis rather than a generic "too small."

When a residual survives every composition of the operators already in the model, that irreducibility is itself the result. It licenses a specific move: *extend the language.* Optimization discovers a residual; the residual, attributed to a morphological axis, generates a hypothesis about the missing physics; the hypothesis is realised as a new registered operator; and the operator enlarges the vocabulary the forward model can express,

$$\text{residual} \longrightarrow \text{hypothesis} \longrightarrow \text{operator} \longrightarrow \text{extended language}.$$

Fitting the data and *building the language that can express it* thereby become the same activity. In the case study below this loop runs to completion more than once. A plateau in loop *enclosure* — trajectories that trace thin, nearly one-dimensional arcs rather than closed loops — is traced not to insufficient force but to a symmetry: a contraction axis that is fixed over the beat retraces its path and encloses no area. The named ingredient is a rotating contraction axis, added as a single operator, which breaks the plateau and fills the enclosure residual. A subsequent plateau in loop *size* then resists every lever the current language offers — drive, gain, stiffness, duration — and prompts a different hypothesis: that the tissue does not enter each beat from a stress-free reference, so that the missing ingredient is a residual-stress (pre-stress) operator, proposed precisely because no operator already in the model can reach it. Each such step both removes a class of models and adds a word to the vocabulary.

The remainder of the paper develops this method and its instantiation: the differentiable active-stress MPM forward model and the operators it is assembled from; the LoopScore morphology metric and why framewise error is the wrong objective for a loop; the residual decomposition that attributes a plateau to a named morphological axis; and the sequence of operator-discovery steps the campaign has produced, presented as the running example of optimization, residual, hypothesis, operator, extended language.